Context or Composition? Explaining the Geographical Polarisation of Cosmopolitan Attitudes in Britain

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Abstract

It is widely accepted that urban populations are more cosmopolitan than rural ones. This article investigates how and why this uneven spatial distribution of attitudes has emerged in Britain, using 21 waves (2016-2025) of the British Election Study Internet Panel and a multi-stage analytical strategy. We find strong evidence that compositional effects drive this patterning but no evidence that contextual effects do the same. Britain's spatial divide over immigration attitudes is therefore primarily a product of self-selection and sorting. More populous places are more cosmopolitan largely because of the distinctive kinds of people who choose to live there and to live elsewhere, rather than because of any transformative effects of local context. Our findings have important implications, showing that geographical divides in British politics reflect deeper demographic and cultural cleavages and cannot be reduced to simple urban-rural binaries, instead requiring more nuanced conceptions of place-based divides.

Introduction

Urban-rural divides in politics are not new; the pitting of industrial and agricultural interests against one another has been shaping party systems, political competition and electoral behaviour since the industrial revolution (Lipset and Rokkan 1967). They have, however, received renewed attention in recent years, as events like Brexit (Abreu and Öner 2020; Becker 2017) and the election of Trump (Albrecht 2022) have highlighted the spatial nature of populism, with the rural vote playing a key role in delivering these outcomes. Scholars have argued that this patterning reflects a 'geography of discontent', whereby decades of urban-focussed growth have created profound divisions between those living in cities, where development and opportunities have been concentrated, and those living in small settlements, who often feel 'left out' or 'left behind', and no longer feel the system 'works for them' (Jennings and Stoker 2017; Mitsch, Lee and Ralph Morrow 2021; Rodríguez-Pose 2018). While the economic explanation for the resurgence of the urban-rural divide is compelling, it is not the only explanation. An alternative (and perhaps complementary) possibility - which has received less attention in the academic literature - is that this divide is cultural, driven by the different political values of urban and rural populations (Mitsch, Lee and Ralph Morrow 2021).

Existing empirical work has shown that cities are bulwarks of cosmopolitan values, with their populations tending to be much more tolerant, liberal and progressive than those of small towns and villages (Hesstvedt and Saglie 2025; Huijsmans et al 2021; Luca et al 2023). However, there is still much to be learned about *why* these patterns of geographic polarisation over attitudes have established themselves. Few studies have sought to explain whether this urban-rural attitudinal divide can be attributed more to compositional or contextual factors (for notable exceptions see Maxwell 2019, 2020) – and none have done so in the British

context[1]. The question of whether the distinctive attitudes of those living in densely populated urban centres is driven more by the experience of living in these kinds of places (contextual effects) or by the self-selection and sorting of specific kinds of people into them (compositional effects) therefore remains a relatively open one.

This study advances our understanding in this area, using data from 21 waves of the British Election Study Internet Panel (BESIP) (Fieldhouse et al 2024) to conduct a multi-stage analysis which explores the extent to which the geographic divide in immigration attitudes observed in Britain from 2015-2026 is driven by compositional and contextual factors. By moving away from the simplistic binary classifications of urban versus rural which have been used to define spatial polarisation in much of the existing work in this area (see Luca et al 2023, Hesstvedt and Saglie 2025 and Maxwell 2019 for notable exceptions) and instead defining geography in a more linear fashion – by focussing on population size or density – it also sheds new light on the precise contours of this divide. Our findings reveal that more populous places in Britain are more supportive of immigration primarily due to the socio-demographic composition of the people who live there; these places are inhabited by a higher proportion of young people, graduates, non-routine workers, ethnic minorities and foreign-born residents. Furthermore, we show that sorting plays a role in this process: people who are open to immigration or have the kinds of socio-demographic characteristics associated with these views are more likely to move to urban areas. In contrast, we find no evidence for contextual effects. Leveraging internal migration to identify the effect of

[1] Though Gallego et al (2016) do assess the relative strength of these influences in determining the voting choices and economic attitudes of people living in different parts of England (as defined by political context) more generally.

residential context on attitudes, we find no evidence that moving to more or less urban locations impacts support for immigration.

This article makes several important contributions. Firstly, in finding that geographic differences over immigration in Britain are driven primarily by compositional effects, this study shows that the same conclusions which have been drawn in alternative contexts (e.g., Maxwell 2019, 2020) also hold in this case. This study thus helps to build consensus around the drivers of attitude formation, and how these processes contribute to geographical polarisation over politics. Secondly, this study confirms that the limited evidence for contextual effects uncovered in existing studies is not a methodological artefact; associated with using sub-optimal two way fixed effects (TWFE) methods to uncover these, rather than causal identification methods, like the PanelMatch procedure (Imai, Kim and Wang 2023), which are less sensitive to model misspecification than regression-based methods (Rauh, Kim and Imai 2025). Thirdly, and finally, in showing that while the relationship between population size and cosmopolitan attitudes is not perfectly linear (with residents of large settlements holding more cosmopolitan attitudes than those of major settlements, on average) it is certainly more linear than many existing studies have implied by using binary urban-rural classifications, this study demonstrates that future studies must adopt more nuanced definitions of the geographical divide over politics if we are to develop a fuller understanding of how and why the population's political preferences cluster in the ways they do.

The Urban-rural Divide in Politics

Geography has been re-discovered as an electoral cleavage in recent years, with a growing body of research identifying and charting the emergence of urban-rural divides (Ford and Jennings 2020). Across a plethora of advanced Western democracies – from the majoritarian

two-party systems of the United Kingdom (UK) and the United States to the proportional multi-party systems of Western Europe – scholars have shown that the inhabitants of large cities tend to think and vote differently to those living in less densely populated areas. The former tend to exhibit more progressive, tolerant and liberal social attitudes, and support green and social democratic parties on account of this, while the latter are typically more socially authoritarian, and thus more inclined to back conservative, populist and nationalist parties (Hesstvedt and Saglie 2025; Huijsmans and Rodden 2024; Huijsmans et al 2021; Maxwell 2020; Scala and Johnson, 2017).

The growing strength of this urban-rural cleavage is a product of the broad societal transformations that have shaped advanced industrialised nations (Huijsmans and Rodden 2024). On the one hand, politics has become increasingly contested along socio-cultural lines since the 1960s. The rise of 'post-materialist' values has driven dramatic increases in the electoral salience of issues including equal opportunities, freedom of expression and environmentalism (Inglehart 1977) – and more recently has manifested in hotly contested debates over immigration and national sovereignty (Dalton 2018; Kriesi et al 2006). This matters because people living in urban centres have been shown to possess starkly different views to people living in small towns and villages on socio-cultural issues. City-dwellers are much more cosmopolitan, tending to be more pro-immigration, pro-European integration, pro equal opportunities and tolerant of 'others' than those living in rural locations (Huijsmans et al 2021; Luca et al 2023; Maxwell 2019, 2020; Jennings and Stoker 2016).

On the other hand, we have globalisation, the rapid expansion of higher education provision, the emergence of the knowledge economy and agglomeration. These changes have meant that secure, well-paid job opportunities are increasingly reserved for those who use their expertise, gained via formal education to degree-level, and 'thinking ability' to complete

their work (Bovens and Wille 2017) – and that these roles are increasingly based in large cities (Moretti 2012; Storper 2018). The combined effect of these forces has been the development of an economic divide between the 'winners', who live in prospering urban centres, and the 'losers', who live in 'left-behind' rural communities (Jennings and Stoker 2017; Rodríguez-Pose 2018), and an opportunity for politicians to mobilise a growing sense of rural 'resentment' and 'consciousness' (Cramer 2016; Trujillo and Crowley 2022; Zollinger 2024).

While there is considerable debate as to the relative contribution made by these social, economic and political forces in driving the geographic polarisation of politics, their combined outcome is clear. Across a whole host of advanced Western democracies, people living in urban centres tend to think and therefore vote differently to people living in rural settlements.

Why Do People Living in Urban Centres Think Differently?

The question that immediately follows is: why is this the case? Existing literature raises two possibilities. The first is a theory of *contextual* effects, which posits that people living in similar places should have common experiences that lead them to develop similar political attitudes (Enos 2017; Johnston and Pattie 2006). People living in large cities may be more cosmopolitan than those living in small towns and villages because they more often encounter people from different neighbourhoods and backgrounds. Larger municipalities tend to have larger shares of foreign residents, religious minority groups and LGBTQ+ persons than smaller settlements. They also tend to have greater population density, meaning that residents are more likely to share space (i.e., on public transport, and in workplaces and schools), which can facilitate meaningful and positive interactions between neighbours, and

thus promote tolerance and understanding of others (Kent 2022; Paluck, Green and Green 2019; Wagner et al 2006). The urban-rural divide in cosmopolitan attitudes may also be a product of urbanites' greater exposure to demographic and economic churn (Maxwell 2020). People living in large cities move more, and urban neighbourhoods change more often – both via demographic shifts and the construction, opening and closing of residential accommodation, businesses and institutions. Experience of these fluctuations could make city-dwellers more comfortable with change and less likely to sacralise traditional culture than rural residents, who are less exposed to this continual churn (Parker 2015; Recchi 2015).

While the contextual effects hypothesis implies that there is something about the experience of living in cities that alters urbanites' attitudes, the compositional effects hypothesis posits that the link between spatial location and political preferences is driven, at least in part, by the fact that individuals with distinctive political orientations disproportionately self-select into urban living (Maxwell 2019, 2020). This 'geographical sorting' has both a socio-demographic and a cultural basis. Practical factors like travel-to-work times and housing costs mean that different kinds of people tend to live in different kinds of places (Gallego et al 2016; Mummolo and Nall 2016). Highly educated individuals in professional occupations are drawn to urban centres, where well-paid knowledge-based occupations have become increasingly concentrated – though they often move to the suburbs as they age and raise families, due to the cost of city living (Elgueta 2024). Those with lower levels of education and who work in manual occupations, who typically have lower-paid roles, however, are increasingly being priced out of cities (Magrini 2019; Partington 2023). These demographic differences could be responsible for the attitudinal divisions between urban and rural spaces because existing research shows that the kinds of people who tend to live in cities – young, highly educated persons in professional occupations – display more cosmopolitan attitudes than the kinds of people who tend to live in smaller settlements –

older, less educated people in manual occupations (Janmaat and Keating 2017; Simon 2022; van de Werfhorst and de Graaf 2004; Zingher 2022). Cultural considerations may also drive sorting. Many large cities are bulwarks of tolerant, liberal and progressive values, known for their open, global outlooks and their celebration of diversity (Luca et al 2023), so it could be that individuals are attracted to these environments by the promise of being surrounded by like-minded individuals, and that those who do not share these values look to move to smaller towns and villages, which typically foster a different cultural milieux (Florida 2003).

Extant research on this question presents something of a puzzle, offering reasons to believe that both contextual and compositional mechanisms drive urban-rural divisions. In direct tests of the relative explanatory power of context and composition, Maxwell (2019, 2020) finds only limited evidence for contextual effects; showing that while lifelong residence in large municipalities does appear to be associated with more cosmopolitan attitudes in Switzerland (though only across some measures of this concept), there is no evidence to suggest that moving to either larger or smaller municipalities is associated with changes in cosmopolitan attitudes in either Switzerland or Germany. The evidence for compositional effects is much more widespread, however. Using cross-national data, Maxwell (2019) finds shows that demographic composition appears to account for much of the urban-rural attitude divide in Europe. Then, using panel data, he shows that demographic and cultural sorting are taking place in Switzerland, though demographic sorting is more consequential when it comes to explaining why cities think differently (Maxwell 2019). Not only are the kinds of people who live in Swiss urban centres disproportionately those who have cosmopolitan attitudes (people with high levels of education and professional occupations) but the people who choose to move to Swiss urban centres are more likely to have cosmopolitan attitudes than non-movers.

However, studies which examine the effects of residential context *in general*, rather than examining these in relation to the rural-urban divide specifically, have found more support for the idea that context matters. In England, Gallego et al (2016) find that movers to safe Conservative seats become more aligned with the local majority, shifting their political preferences in an economically right-wing direction. Furthermore, research has shown that living in parts of France, Germany and the Netherlands characterised by larger immigrant populations leads residents to develop more favourable attitudes toward immigration and ethnic minorities and reduces support for far-right parties (Achard et al 2025; Jolly and DiGuisto 2014; Stolle et al 2013; Vasilopoulos, McAvay and Brouard 2022). Contextual effects of the opposite form have also been detected though; with Ajzenman, Aksoy and Guriev (2022) and Igarashi and Laurence (2021) showing that greater exposure to non-native populations creates a 'threat' to the native population and serves to push their attitudes in an anti-immigration direction. Whilst the direction of contextual effects appears up for debate, this literature emphasises that the environments in which people live have important effects in shaping the way they think.

This body of evidence raises questions about whether contextual or compositional effects best explain the urban-rural attitudinal divide. Whilst concurrent tests of the two mechanisms suggest that composition matters more when it comes to explaining the urban concentration of cosmopolitan attitudes (Maxwell 2019, 2020), other studies have provided robust longitudinal evidence to suggest that context shapes our political preferences (e.g., Gallego et al 2016; Vasilopoulos, McAvay and Brouard 2022). Given this puzzle, further research into this topic – across different advanced Western democratic contexts and using different measures of this concept – is necessary to draw more robust and well-founded conclusions about *why* the urban-rural divide in politics has emerged, and the relative contribution of compositional and contextual effects in this process.

This study takes up precisely this task. Doing so not only promises to address a gap in the literature, by advancing our understanding of why different kinds of places have come to think and vote differently, but also to produce findings that have important real-world implications. Living in highly geographically polarised societies poses risks – heightened animosity between the 'winners' and 'losers' of globalisation can reduce social cohesion, and lead to situations of political deadlock as politicians pit urban interests against rural ones. Only by understanding how these place-based divides have come to be can we hope to understand how they might be reconciled in the future.

We make an additional contribution to the literature by moving beyond the stark urban-rural binary which has been the focus of much existing research on this topic. While several studies have shown that the attitudes and voting patterns of people living in large urban centres differ from those living in small towns and villages (e.g., Huisjmans and Rodden 2024; Jennings and Stoker 2016) – and begun to explain *why* these patterns of geographical polarisation over politics occur – they tell us little about whether there may be a more linear relationship between settlement size and political preferences. This is problematic, given firstly, that Luca et al (2023), Hesstvedt and Saglie (2025) and Maxwell (2019) have all found evidence to suggest that the geographical patterning of political preferences extends beyond a simple urban-rural divide, and secondly, given that there seems no reason, in light of the mechanisms that link spatial location with political preferences, to expect that living in medium sized settlements would not have a distinctive effect on attitude formation, as compared to living in large and small settlements. This study will put this assumption to the test, and in doing so, considerably advance our understanding of *how* and *why* political values are unevenly distributed across space.

Research Questions

This study aims to shed light on *how* and *why* settlement size is associated with aggregate political values. It will start by asking: *do individuals residing in more populous settlements have more cosmopolitan attitudes than those in less populous settlements? [RQ1].* Assuming that the answer to this question is yes, as the existing literature (e.g., Huijsmans et al 2021; Jennings and Stoker 2016; Luca et al 2023; Maxwell 2019, 2020) suggests it should be, this study will then go on to establish the extent to which contextual and compositional effects can explain this uneven distribution of attitudes across space. To address the potential for contextual effects, we ask: *does moving to a more (less) populous area lead to the adoption of more (less) cosmpolitan attitudes? [RQ2].* To consider whether compositional effects are at play, we ask the following three questions: *does the socio-demographic make-up of more and less populous places differ? [RQ3] Can this explain the geographic divide in attitudes (i.e., do the groups over-represented in larger (smaller) settlements have more (less) cosmopolitan attitudes?) [RQ4] And do we find evidence of demographic and attitudinal sorting, whereby people who have, or are likely to have, cosmopolitan attitudes are more likely to move to more populous settlements and vice versa? [RQ5].*

Following Maxwell (2019, 2020) we define cosmopolitanism primarily in terms of attitudes toward immigration, though we also employ an alternative measure which captures attitudes towards the European Union (EU) in the supplementary materials. Our findings are substantively similar across specifications.

Britain as a Case

The British context is an ideal one for study, because the political preferences of city and country have long been starkly divided in Britain. Since the 19th century, the Conservative Party have held a pronounced rural electoral advantage over their main rivals (Claassen 2024; Lipset and Rokkan 1967). More recently, Remain support at the 2016 EU referendum was heavily concentrated in large cities and university towns whilst Leave support was clustered in 'left behind' towns, declining coastal areas and the countryside (Abreu and Öner 2020; Becker 2017) – a divide which was based, at least in part, on the different kinds of attitudes held by the people living in these places (Jennings and Stoker 2016). Britain is also a place where the structural changes that have given rise to geographic polarisation (e.g., globalisation and agglomeration) have been felt particularly profoundly (McCann 2016, 2020) and one where there is a high degree of residential mobility compared to other developed nations (Causa and Pichelmann 2020).

The salience of urban-rural divides in Britain during the period we study makes our context something of a 'most likely' case to identify contextual effects. In tandem, high residential mobility in Britain provides us with the necessary variation in residential context for observing the processes of interest in RQ2. Testing the contextual effects hypothesis in this context thus helps us to understand whether the weak evidence identified in previous research (Maxwell 2019, 2020) is a product of the places and periods under study.

Data

We use data from the high-quality, large-N (~30,000 respondents per wave) BESIP (Fieldhouse et al 2024), as its regular wave structure (minimum one per year) and long

duration (2014-2026) provides both the statistical power and span of data collection that is necessary for analysing the processes that produce geographical polarisation in political attitudes.

We take local authorities as our unit of spatial analysis in much of what follows.[2] In Britain, local authorities are electoral and administrative units responsible for delivering public services, not spatial boundaries that accurately reflect the population density of the places people live, work and socialise. Rather than simply using local authority population figures to differentiate between the 'size' of these units, we therefore combine this information with that on an alternative statistical geography – the built-up area. These boundaries, created by the UK Office for National Statistics (ONS), are designed to capture the precise contours of the villages, towns and cities in England and Wales in which people spend their time, and thus facilitate more accurate comparisons between the 'sizes' of the different kinds of places in which people actually live their lives (ONS Geography, 2015). We combine information on these geographical units by calculating the weighted average of the usual resident population across all built-up areas sitting within each local authority boundary and down weighting the figures for all built-up areas sitting only partially within these boundaries.[3] Through this process, we determine the size of the average settlement that sits within (and across) each local authority's boundaries. This provides us with a population density measure that better captures the 'size' of the places where people spend their time.

[2] The exception is our analysis of contextual effects, where we use middle-layer super output areas (MSOAs), as the smaller scale of these spatial units is more commensurate with the environments in which we interact socially.

[3] For a detailed description of this process and information on local authority population averages – including a map of English and Welsh local authorities by population size - see supplementary materials section 1.

We use the median population figure for each local authority in our analysis, due to its reduced sensitivity to outliers (relative to the mean).

The Geographic Structuring of Immigration Attitudes in Britain

We begin by testing the assumption that individuals residing in more populous parts of Britain are more supportive of immigration than those living in less populous localities. We do this by comparing average scores on our outcome measure across local authorities of different settlement sizes, where settlements are classed as follows, based on their average population figures: minor (<10,000); small (10,000-99,999); medium (100,000-499,999); large (500,000-999,999); and major (>1,000,000). Our outcome variable is an 11-point (0-10) scale measure of immigration attitudes, which asks respondents whether 'the UK should allow many more immigrants to come to the UK to live' or instead 'allow many fewer immigrants'. Higher scores indicate greater support for immigration. This question is first asked at BESIP wave 7 (April 2016) and then at every subsequent wave except for 18, 19 and 24. Our models therefore use 21 waves of BESIP data covering the period 2016-2025.

To compare attitudes across local authority types, we first calculate pooled mean scores for our measure of support for immigration across all BESIP waves (see Table 1). Our calculations show firstly that, on average, individuals living in all types of local authority wish to reduce immigration to Britain (indicated by scores below 5). Secondly, they point to differences between local authority types. Individuals residing in large and major settlements are considerably more open to immigration than those residing in the other three 'smaller' locality types, which all exhibit similar average scores. This trend is consistent with findings from prior research which indicate that geographic differences in socio-cultural attitudes are primarily divided between densely populated urban areas and elsewhere.

Table 1: Immigration Attitudes by Locality Type (Pooled Mean Scores)

	Locality Type				
	Minor	Small	Medium	Large	Major
Immigration Attitudes	3.37 (2.82)	3.51 (2.88)	3.4 (2.89)	4.18 (3.00)	3.96 (2.98)

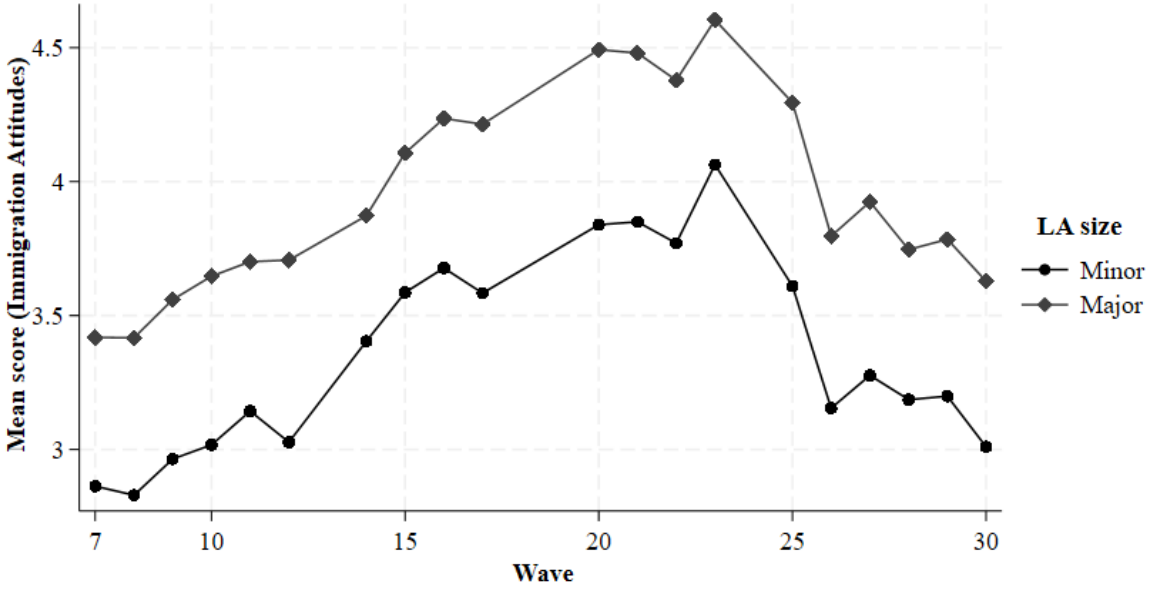
Note: higher scores indicate greater openness to immigration. Mean scores across panel observations are shown with standard deviations in parentheses.

Next, we calculate time trends in average immigration attitudes across local authority types. Due to concerns with low *N* at individual waves within small, medium and large local authorities, we only plot wave-specific means for those individuals residing in major and minor local authorities, which together constitute 80% of person-observations in the data. Figure 1 indicates clear attitudinal differences between the two local authority types, with the populations of major local authorities being more pro-immigration than those of minor local authorities, on average, across every BESIP wave studied.

In sum, comparisons of political attitudes between locality types clearly indicate that individuals living in more populous areas exhibit more pro-immigration attitudes than those in less populous areas. Figure 1 also shows that attitudes are not static. This is unsurprising given the political shocks and party leadership changes that occurred over the study period (Fieldhouse et al 2020). Importantly however, these changes in attitudes over time follow similar trends across geographies, meaning that the relative difference between the attitudes of people living in more urban and more rural areas remains broadly constant.

The existence of these differences raises the question of what drives them. Do different kinds of people, with different kinds of immigration attitudes, simply choose to live in different kinds of places (i.e., compositional effects)? Or is it the experience of living in different kinds of places that leads the people living there to develop different attitudes towards immigration (i.e., contextual effects)? The following sections unpack these competing possibilities.

Figure 1: Average Immigration Attitudes in Minor and Major Local Authorities, Across BESIP Waves



The y-axis plots wave-specific mean scores on the our measure of openness to immigration in each local authority type, coded so that higher scores equate to greater openness. Only waves with 1000 or more observations for each wave-category combination are plotted.

Testing the Compositional Effects Hypothesis

To what extent is the greater tolerance of immigration observed in more populous vis-à-vis less populous parts of Britain driven by differences in the kinds of people who inhabit them? If compositional effects are at play, not only will the socio-demographic make-up of these areas differ, but the same socio-demographics that make the populations of these places distinctive will also be associated with immigration attitudes. Here, we test both possibilities. We then go further by testing 'sorting' as a driver of compositional effects, considering whether our demographic characteristics and political attitudes predict the kinds of location we choose to move into and out of.

Using local authority-level data from the 2021 Census of England and Wales, we begin by calculating the average (mean) proportion of the population with specific age,

education, occupation and ethnicity-based socio-demographic characteristics across our five local authority types (see Table 2). We find clear and distinct socio-demographic differences between settlements of different sizes. The populations of larger settlements are, on average, much younger and more diverse (both ethnically, and in terms of their place of birth) than those of smaller settlements. Residents of major settlements are also considerably more educated, on average, than residents of other settlement types. Occupation-based differences are less clear cut, however. While fairly similar shares of working-class persons (those in routine and semi-routine work) are found in all settlement types, except major settlements, where there are somewhat less working-class persons, the highest shares of middle-class persons (managers and professionals) are found in minor and major settlements, with relatively fewer middle-class persons living in 'middling sized' settlements.

Table 2: Socio-demographic Characteristics by Locality Type

Population characteristics	**Locality Type**				
	Minor	Small	Medium	Large	Major
Aged 16-34	21.05	23.32	25.43	31.43	28.19
Aged 65+	21.85	18.83	18.04	14.51	13.23
White	91.77	89.50	84.61	71.60	61.00
UK-born	90.32	89.30	84.28	77.17	66.03
Degree-educated	32.09	30.70	29.66	31.71	42.10
(Semi-)Routine occupations	23.70	25.37	26.21	25.18	19.54
Managers and professionals	34.07	31.60	29.44	27.07	35.99

Note: degree-educated are those with at least level 4 qualifications; managers and professionals are defined using the NS-SeC occupational classification system. This data is drawn from the 2021 Census and uses 2023 boundaries.

Nevertheless, it is clear that more and less populous parts of Britain tend to be populated by different kinds of people; with the former more often being home to younger, degree-educated, ethnically diverse individuals and persons that have not been born in the UK than the latter. If individuals with these characteristics, who tend to be over-represented in urban

areas (as shown in Table 2), also tend to have more pro-immigration attitudes, then the unique sociodemographic composition of cities and other densely populated metropolitan centres may well explain why their aggregate-level attitudes differ to those of less populous places.

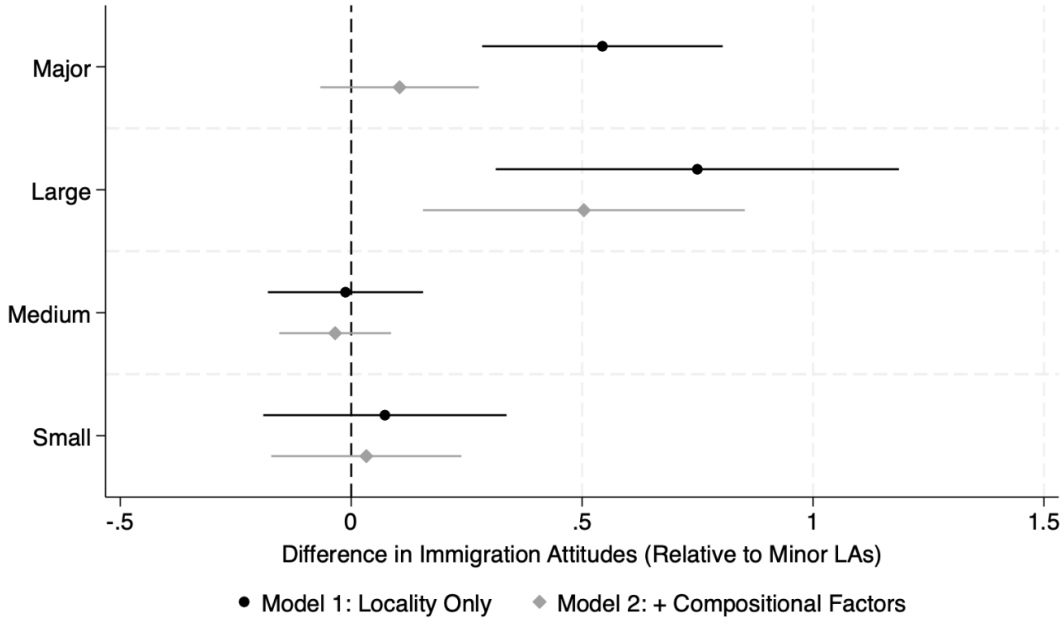
We test this possibility with a two-step analysis. First, we regress immigration attitudes on our five-category local authority variable, to test whether individuals who reside in more heavily populated settlements have different attitudes to those in less populous areas. Then, we add socio-demographic characteristics of interest to this model, to measure the compositional differences between different local authority types. Specifically, we add age, ethnicity (a dummy for whether the respondent is white), education (a dummy for degree education), immigration status (a dummy indicating whether the respondent is UK born) and occupational class (routine/semi routine, managerial and professional, intermediate, small employers and lower supervisory/technical) in this second model. If the differences between immigration attitudes observed between local authority types reduce in magnitude after these socio-demographic factors are included as covariates, then we have evidence to suggest that compositional effects drive these differences.

As we use panel data, and our models include multiple observations per individual, we cluster standard errors by individual and local authority. All models also include wave fixed-effects to account for time-varying factors – like national economic conditions or elite behaviour – that might impact immigration attitudes, and are estimated on identical samples, to ensure that any differences observed between models are the result of including socio-demographic controls, rather than changing sample composition.

Figure 2 compares the locality coefficients between models that include (Model 2) and exclude (Model 1) compositional controls (see supplementary materials Table 2.1 for full

output of both models). It provides clear evidence in favour of compositional effects. Model 1 indicates that, relative to those residing in minor settlements, individuals living in major and large settlements are considerably more supportive of immigration. When socio-demographic characteristics are included in Model 2 however, the magnitude of these geographical divisions diminish. The difference in immigration attitudes between minor and major settlements disappears, and the difference between minor and large settlements reduces in magnitude but remains statistically significant.

Figure 2: The Effect of Locality Type on Immigration Attitudes, Before and After Socio-Demographic (Compositional) Controls



Note: Results are OLS regression coefficients representing the difference in immigration attitudes between individuals residing in the corresponding locality type and those living in minor local authorities, with robust 95% CIs (clustered by individual and local authority). Model N = 341,437

Furthermore – as shown in supplementary materials Table 2.1 – age, ethnicity, country of birth, education and occupational class all predict immigration attitudes. Consistent with the logic of compositional effects, the same kinds of people who are over-represented in more populous areas vis-à-vis less populous ones, also tend to be those who are more pro-

immigration. This suggests that the different socio-demographic compositions of these places in Britain explains a large portion of the attitudinal differences observed between them.

Examining Sorting

Compositional effects may be driven by 'sorting'. If the demographic groups that favour immigration are more likely to move to more populous areas, then processes of internal migration will contribute to the urban-rural attitudinal divide (socio-demographic sorting). This may then be exacerbated by a process of cultural sorting, if people select to live in certain kinds of areas based on pre-existing attitudes (i.e., choosing to live in areas where others share their attitudinal preferences and avoiding those where they do not).

If socio-demographic and cultural sorting were taking place, then individual-level attitudes and socio-demographics characteristics would predict the likelihood of moving into and out of different kinds of areas. Specifically, some or all of being pro-immigration, young, non-white, non-UK born, degree-educated, and non-routine occupations would be expected to predict moving to more populous settlements (with the opposite holding true for less populous settlements). We test this possibility by exploring whether the propensity of individuals to move to more and less populous settlements is predicted by these characteristics.

Age is highly associated with residential mobility, with younger people moving more frequently than older people as they leave the parental home to seek new opportunities and form their own families (Ministry of Housing Community and Local Government 2026). Education and occupation may shape residential mobility too, with movers tending to be more advantaged (Playford, Mountford-Zimdars and Harrison 2026). To disaggregate

processes of socio-demographic sorting from more general patterns of residential mobility, we restrict our analysis to individuals who relocate to a different local authority (resulting in a between-mover comparison), rather than comparing movers with non-movers. Specifically, we assess whether attitudes and socio-demographic characteristics prior to moving are associated with the *type* of move that an individual undertakes.

The outcome variable here is the (beta-standardized) population size of the local authority that each mover relocates to. This is measured continuously – as number of persons – rather than categorically – in terms of population size groupings, as previously.[4] We regress this measure on immigration attitudes and socio-demographic characteristics, all measured at T_{-1} (i.e., the measurement occasion immediately prior to moving). This allows us to determine whether an individual's immigration attitudes and socio-demographic characteristics, prior to moving, influence their choice of environment to move to – providing a test of both cultural and socio-demographic sorting. We use lagged versions of these measures as we are interested in the impact of pre-move attitudes and demographics on moving and post-move measures are potentially endogenous to the individual's new residential context. We also include a measure of population size at T_{-1} as a control (i.e., a lagged dependent variable, recording the size of the local authority each mover lived in previously). If residential self-selection is occurring, the characteristics of the area being moved from are likely to be central to both the decision to leave and the kind of area that is moved to. In other words, population size at T_{-1} will be an important determinant of the decision to move and the choice of new

[4] In supplementary materials table 2.2 we replicate this analysis with the categorical local authority indicator as the outcome measure in an ordered logistic regression. Results are largely consistent across specifications.

location (i.e., treatment assignment). We also include wave fixed effects to capture time trends, and cluster standard errors by individual and post-move local authority.[5]

Estimates from this model are presented in Table 3 and provide evidence of both socio-demographic and cultural sorting. Not only do we find that individuals who are more supportive of immigration are more likely to move to more heavily populated contexts, even after socio-demographic controls, but also that older (younger) individuals are less likely to move to less (more) heavily populated areas, and that white individuals more often move to less heavily populated areas. The geographical divide over immigration attitudes in Britain is produced both by a cultural sorting mechanism and a process of socio-demographic sorting, in which younger and non-white individuals - who are more likely to possess liberal attitudes toward immigration – more often move to more urban areas, and older and white individuals – who are more likely to oppose immigration – more often make the opposite kind of move.

Table 3: The Effect of Socio-demographic Characteristics and Immigration Attitudes on 'Move Type'

Variables	**Post-move Population**
Immigration attitudes T_{-1}	0.016* (0.006)
Age T_{-1}	-0.007*** (0.001)
White T_{-1}	-0.211*** (0.061)
UK Born T_{-1}	-0.017 (0.051)
Degree-educated T_{-1}	0.063* (0.029)
Managers and Professionals T_{-1}	-0.019 (0.035)
Intermediate T_{-1}	-0.015 (0.040)
Small employers T_{-1}	-0.016 (0.062)

[5] Most respondents in this model appear once, but individuals who move multiple times are observed on repeated occasions. We thus cluster standard errors by individual as well as local authority.

Variables	Post-move Population
Lower supervisory/technical T_{-1}	-0.043 (0.057)
LA population (lagged) T_{-1}	0.388*** (0.042)
Wave FE	YES
Observations	5002

*Note: Results are OLS regression coefficients, with robust standard errors (clustered at the individual and local authority level) in parentheses. *** p<0.001, ** p<0.01, * p<0.05.*

Testing the Contextual Effects Hypothesis

While the analysis presented previously makes evident that compositional effects go some way in explaining the spatial patterning of immigration attitudes in Britain, the door remains open for contextual effects. Even after controlling for the different kinds of people who tend to live in these different kinds of areas, urban residents (those living in large local authorities) remained substantially more pro-immigration than rural residents (those living in minor local authorities) (see Figure 2). Could there something about the day-to-day experience of living in more populous places that causes people to become more supportive of immigration?

To examine the validity of the contextual effects hypothesis, we analyse the impact of changing residential location on immigration attitudes. Not only are individuals residing in more populous areas more supportive of immigration on average than those residing in less populous places (see Table 1 and Figures 1 and 2); individuals with pro-immigration attitudes are also more likely to self-select into city living (Table 3). As are young persons, non-white people, and people with degree-level qualifications, who are more often supportive of immigration (Table 3), and are over-represented in densely populated urban centres as compared to more sparsely populated rural parts of Britain (see Table 2). Could it be that moving to a more (less) populous part of Britain where you are more (less) likely to rub

shoulders and/or interact with these kinds of people produces an increase (decrease) in support for immigration?

If contextual effects are in operation, they are likely to occur via social interaction (Sharkey and Faber 2014). A key site for interaction is the neighbourhood in which one lives (and moves to).[6] Given the need for fit between theory and measurement in the estimation of geographic effects (e.g. Lee, Rogers and Soifer 2025), we avoid using local authorities as our unit of geography in this analysis. Local authorities can be very large, with metropolitan authorities like Birmingham containing over 1 million residents (ONS Demography, 2026). This means that many local authorities are heterogenous, containing areas that are diverse and densely populated, along with areas (such as suburbs) that are more ethnically and culturally homogenous and contain less of the churn that defines urban centres. Given this, we instead use an alternative geography – the MSOA – to define moves here. MSOAs are census units used in England and Wales which contain 2000-6000 households. Their smaller size means they more effectively capture the neighbourhoods in which people live.

As MSOAs are designed to have similar population sizes, we use population density to distinguish between more and less populous neighbourhoods. We divide annual mid-year MSOA population estimates by the geographic area of each MSOA to calculate mid-year population density estimates for each from 2014-2025. We then assign each BESIP wave to the closest mid-year estimate it is fielded after. Within each wave, we identify the percentile ranking (0-100) of each MSOA based on population density. In our analysis, we focus on the effect of moving to both higher-density and lower-density MSOAs. We define a person-

[6] Whilst the neighbourhood is not the only context in which interaction takes place – the school and workplace are other important examples (Bohman and Miklikowska 2021; Mutz and Mondak 2006) – we focus on neighbourhood context due to the data and measures we have available.

observation at wave T as a move to a higher-density MSOA when the individual resided in a different MSOA with a lower density percentile at T_{-1}. Moves to lower-density MSOAs are defined as person-observations at T when the individual resided in a different MSOA with a higher density percentile at T_{-1}.

If the contextual effects hypothesis holds, we would expect that moving to higher-density MSOAs would have a positive effect on support for immigration, whereas moving to lower-density MSOAs would have a negative effect. To test this, we use the PanelMatch estimator (Imai, Kim and Wang 2023). PanelMatch provides an extension of typical (cross-sectional) matching estimators for panel data, matching treated units with control units with similar covariate histories to construct a transparent counterfactual, before estimating treatment effects using a difference-in-differences estimator.

Given that the mechanisms by which immigrants and non-immigrants form their attitudes toward immigration likely differ (Šedovič and Dražanová 2025), we restrict our sample to UK-born respondents. Our two causal estimands are the average treatment effect among higher-density and lower-density movers (the ATT), or the average difference in post-move immigration attitudes between movers and their counterfactual outcome if they had not moved. Causal identification rests on the assumption of (conditional) parallel trends between treated units (movers) and control units (non-movers). To increase the plausibility of this assumption, we match treated and control units on several dimensions.

We begin by constructing unrefined candidate sets for treated units using 4-wave lags. In other words, we match eventually treated units with control units that share treatment histories for 4 waves prior to treatment. We use 4-wave lags as we deem it to provide the best trade-off between (a) accounting for potential pre-trends by balancing on pre-treatment covariates and (b) statistical power (the number of valid treated and control units decrease as

we use longer lags due to missing data). This results in 1028 treated units who move to higher-density MSOAs and 1225 treated units who move to lower-density MSOAs. By matching on waves and treatment history, we account for common (i.e., national-level) time trends or wave-specific shocks and ensure our treatment estimates are not biased by prior moves within this four-wave window. The candidate set for each treated unit is large, with no empty sets (i.e., no treated unit lacks a potential control). The median candidate set for higher-density treated units contains 1755 control units (minimum = 150), whilst the median candidate set for lower-density treated units is 2153 (minimum = 147). We then refine the candidate sets for each treated unit by balancing on a series of observables (i.e., matching treated units with control units that possess similar histories on potential confounders).

Specifically, we match on ethnicity, age, education and occupation (from T_0 to T_{-4}), and pre-treatment population density and lagged immigration attitudes (from T_{-1} to T_{-4}). We use three refinement procedures to achieve covariate balance: Mahalanobis distance matching, propensity score matching and covariate-balancing propensity score (CBPS) matching. Matching on socio-demographics reduces concern that our estimates are biased by socio-demographic sorting, and matching on pre-treatment population density helps us rule out confounding via past residential context. Finally, matching on lagged immigration attitudes ensures balance on the outcome prior to treatment – strengthening the assumption of parallel trends – and reduces concern with selection into treatment via cultural sorting (Cho, Gimpel and Hui 2013; Gallego et al 2016).

This approach provides several benefits over the linear TWFE (i.e., unit and time fixed effects) that have typically been used when estimating the effect of changing residential location on political preferences (e.g., Maxwell 2019, 2020). First, we construct a clear and defined counterfactual by comparing treated units with control units that share the same prior

treatment history (for up to 4 waves). Second, by conditioning on lagged treatment and outcomes, we are better able to account for the existence of dynamic relationships between treatment and outcome (Imai and Kim 2019). Third, we avoid the attenuation bias that can be produced by using unit and time fixed effects (Imai and Kim 2021).

Figure 3a and 3b display balance scores across covariates for the unrefined and refined matched sets. They show that, as we would expect, refinement improves balance on covariates. Using ± 0.2 standard deviations as the cut-off for acceptable balance (as suggested by Rauh, Kim and Imai 2025), Mahalanobis and CBPS matching achieves acceptable balance on all covariates for both moves to higher-density and lower-density MSOAs. Given these balance scores, we estimate the ATT for moving to higher-density and lower-density MSOAs from the Mahalanobis-refined and CBPS-refined matched sets, with bootstrap standard errors (1000 repetitions). Results are presented in Table 4 and show no evidence that either 'move type' exerts an effect on immigration attitudes – confidence intervals are large and include 0 for all treatment estimates.

Figure 3a: Balance Scores, Moving to Higher-Density MSOAs

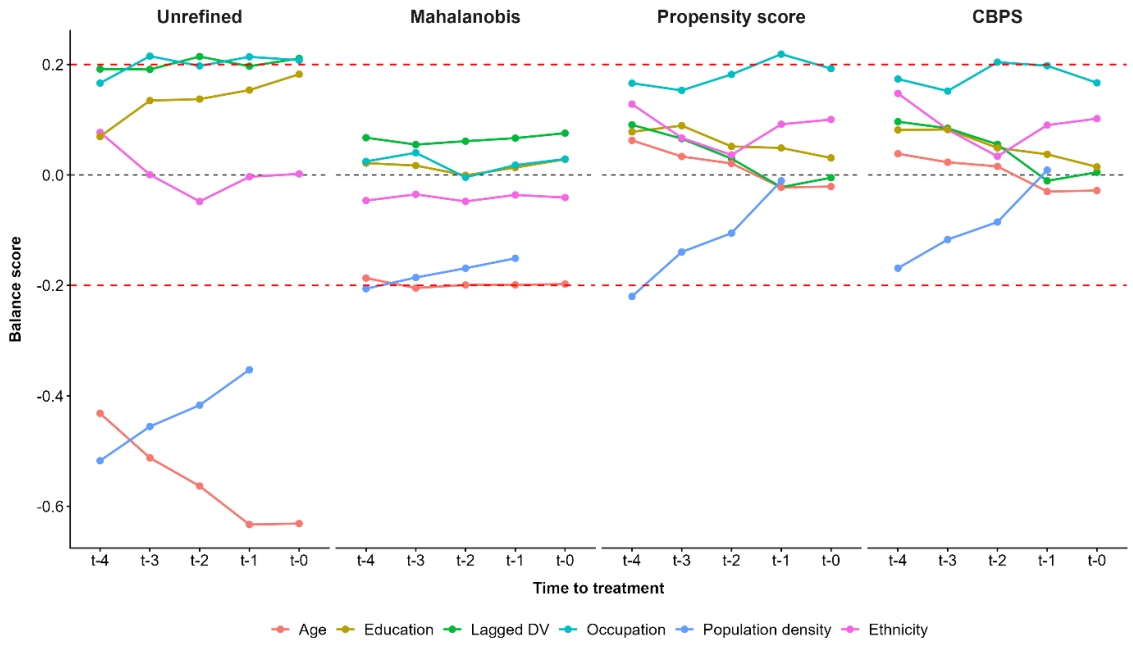


Figure 3b: Balance Scores, Moving to Lower-Density MSOAs

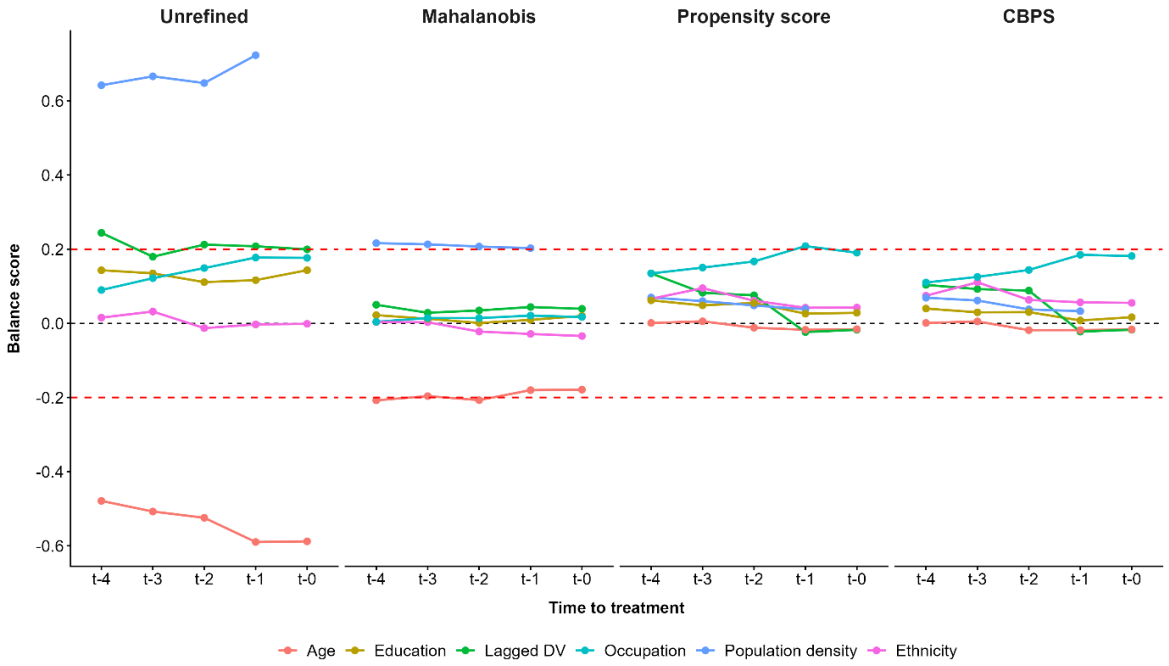


Table 4: ATTs for Moving to Higher-density and Lower-density MSOAs on Immigration Attitudes

Matching Procedure	**Moving to Higher-density MSOA**				**Moving to Lower-density MSOA**			
	ATT	SE	Lower CI	Upper CI	ATT	SE	Lower CI	Upper CI
Mahalanobis	0.028	0.057	-0.081	0.136	-0.015	0.052	-0.117	0.08
CBPS	0.045	0.061	-0.076	0.164	0.015	0.055	-0.097	0.119

Robustness Tests

Our PanelMatch estimates provide no evidence of a contextual effect of changing residential context on immigration attitudes. In supplementary materials section 3, we conduct several additional tests, to ensure this null result is genuine. Firstly, we estimate models that capture

only moves that come with large changes in MSOA population density (see supplementary material section 3.1), this helps to assess whether our estimates in Table 4 are biased downwards by including moves that are associated with only minor population density changes, which may have smaller effects on our attitudes. Secondly, we estimate an alternative version of the PanelMatch procedure that does not match on T_0 occupation and education, as matching on contemporaneous occupation and education could introduce post-treatment bias into our estimates (see supplementary material section 3.2). Thirdly, and informed by the 'impressionable years' hypothesis (Sears and Funk 1999), we test whether treatment effects exist only among respondents who are 30 or younger and thus have more malleable attitudes (see supplementary materials section 3.3).

We also test whether our results differ when we use preferences for EU integration as our outcome variable and define our treatment in terms of changing local authority population characteristics (see supplementary materials sections 3.4 and 3.5 respectively). Finally, we model the effect of changing residential context using an alternative estimation strategy, helping us to validate that our results are not the artefact of a specific modelling strategy (see supplementary materials section 3.6). Specifically, we adopt the TWFE strategy used in previous tests of contextual effects via changes in residential context (Maxwell 2019, 2020). Results across all robustness tests also fail to support the hypothesis of contextual effects; we find no evidence that the experience of moving to either higher-density or lower-density residential contexts impacts our attitudes – regardless of the methods or measures used when trying to uncover these effects.

Discussion and Conclusion

This paper goes beyond the scope of existing work not only by considering the extent to which and reasons why cosmopolitan attitudes are unevenly distributed across space in Britain, but by considering whether this patterning is more nuanced than traditional conceptualisations of urban-rural divides would suggest. In doing so, it advances our understanding of the relative explanatory power of context and composition-based accounts of the geographical polarisation of political values – revealing that, in the British case, as in other European contexts, self-selection and sorting are the primary mechanisms driving this patterning. We find strong evidence of compositional effects. The more cosmopolitan attitudes of more populous settlements are largely a product of the distinctive demographic characteristics and ways of thinking of the people who choose to live in and migrate into and out of these places – though this account carries more power when it comes to explaining the attitudinal differences observed between major and minor settlements, than it does the differences between large and minor settlements. On the other hand, we find no evidence of contextual effects. Neither moving to more or less populous settlements has any discernible effect on immigration attitudes in Britain, and this null effect is robust to alternative specifications. The remainder of this section discusses the implications of our findings.

This study's findings support and extend the evidence base which underpins the emerging consensus in the literature on spatial polarisation over politics, which holds that compositional effects are the most important drivers of this patterning. While existing studies have found this to be case for cosmopolitan attitudes in Europe (Maxwell 2019, 2020) and for economic attitudes and vote choice in Britain (Gallego et al 2016), our study shows that the same conclusion holds more widely, also being uncovered when considering the former outcome variable (cosmopolitan attitudes) in the latter context (Britain). We find extensive

support for composition-based explanations, including evidence that both socio-demographic and cultural sorting contribute to shaping the distinctive attitudinal character of more populus settlements vis-à-vis less populus ones. Not only do we find that people who are more supportive of immigration are more likely to move to more populous settlements (even after socio-demographic controls) and that younger and non-white individuals – who are more likely to possess liberal attitudes toward immigration – more often move to more urban areas, and older and white individuals – who are more likely to oppose immigration – more often make the opposite kind of move, but we also find that major settlements are no longer statistically significantly more pro-immigration than minor settlements once we control for distinctive characteristics of the populations of these places.

What our study cannot determine, however, is what motivates these moves. Like Cho, Gimpel and Hui (2023), we find that where people move, they tend to relocate to places with populations that think more 'like themselves' when it comes to politics, but we cannot identify whether this pattern is driven by a genuine desire to live among like-minded individuals or whether it is simply a product of the fact that structural constraints – like affordability and travel-to-work times – shape the choices that different socio-demographic groups can make about where they live (Mummolo and Nall 2016). Given that the same groups which are increasingly being priced out of urban centres, including people who work in manual occupations and have lower levels of education (Magrini 2019; Partington 2023) also tend to be those with less cosmopolitan attitudes (e.g., Bovens and Wille 2017; Janmaat and Keating 2017; Simon 2022; van de Werfhorst and de Graaf 2004; Zingher 2022), the latter explanation could certainly be plausible. While seeking to uncover whether the patterns of cultural sorting identified in this study are genuine or simply a proxy for socio-demographic sorting is an important avenue for future research, the central claim of this study – that different kinds of places think in the ways they do largely because of the distinctive

kinds of people who choose to live there (or to live elsewhere) – holds regardless of the motivation that underpins this 'moving behaviour'. Like Maxwell (2019) we find clear evidence to suggest that the geographical polarisation of politics is a second-order manifestation of deeper demographic and cultural divides.

This conclusion is further underscored by the fact we find no evidence of *contextual* effects in our study. Just as Maxwell (2019, 2020) demonstrates in the Swiss and German contexts, we show that neither the experience of moving to more or less populous settlements has any effect, on average, on the immigration attitudes of people living in Britain. Robustness checks presented in the supplementary materials confirm that these null findings are not simply a product of including individuals making 'insignificant' moves in our 'treatment group' or analysing a sample that does not have particularly malleable attitudes, and that they hold regardless of the specific outcome measure and geographical units our analysis employs. They also confirm that our findings do not differ substantively when we estimate our contextual effects models using TWFE rather than PanelMatch. This is perhaps surprising. We anticipated that using the PanelMatch methodology which is 'more robust against model misspecification compared to regression-based approaches' (Rauh, Kim and Imia 2025: 2), may allow us to detect small contextual effects where TWFE approaches could not, should they genuinely exist. That we did not may be taken as evidence that contextual effects do not constitute an important part of this story. Given the ease of implementing the PanelMatch procedure in R, and that it may produce different results to the TWFE method – which has become the 'default choice' for many seeking to estimate contextual effects – we recommend that future studies seeking to explain the geographical polarisation of political values implement this more sophisticated methodology, at least as a robustness check.

It is worth noting, however, that Maxwell (2020) does find some evidence of contextual effects in Switzerland when lifelong residence is considered; with those who have

lived in larger municipalities for their entire lifetime being more cosmopolitan than those who have always lived in smaller towns and villages. Could our null finding simply be a product of not having observed individuals for long enough post-move to discern context-based effects then? The overall duration of the BESIP and the span of occasions at which our key outcome measure of immigration attitudes was fielded constrains the period over which we can observe attitudinal changes related to residential moves – meaning that, at most, we can track individuals over 9 years from 2016-2025. We argue that this should not preclude our ability to detect context effects should they exist though, as Gallego et al's (2016) analysis finds evidence that people living in England begin to assimilate, taking on the political preferences of those in the 'new' constituencies they have moved to, as early as 2-3 years post-move.

Though our study finds no concrete evidence of *contextual* effects, we are unable to explain the tendency of more populous places to have more pro-immigration attitudes than less populous places through *compositional* factors alone. While we find that the populations of both large and major settlements have statistically significantly more cosmopolitan attitudes compared to those of minor settlements, on average, only the attitudinal differences between major and minor settlements are nullified after controlling for the differing socio-demographic compositions of these places (see Figure 2). The size of the 'attitudinal gap' observed between large and minor settlements is reduced considerably by the introduction of these controls but nevertheless remains substantial and statistically significant. There are two potential explanations for this finding. It may be that compositional factors are the only ones that matter in driving the geographical polarisation of attitudes in Britain, but that we have omitted vital measures of population composition from our analysis – which if included, would 'explain away' this remaining attitudinal disparity. We think this unlikely, however, as our study includes the full suite of relevant compositional variables identified in existing

works on the determinants of geographical divides over politics (e.g., Gallego et al 2016; Huijsmans et al 2021; Luca et al 2023; Maxwell 2019, 2020). That said, many of these studies also come to the same conclusion – that although compositional effects are strong, they cannot explain the spatial divide in attitudes away in its entirety (e.g., Huijsmans et al 2021; Luca et al 2023; Maxwell 2019). Perhaps then, the alternative explanation, which is that contextual effects do play at least some small role in determining the geographical clustering of attitudes is more plausible.

This highlights an important limitation of our study. It is possible that our strategy of searching for 'average' shifts in the immigration attitudes of those who have moved to more and less populous settlements obscures important differences in these effects between sub-groups. Harteveld and van der Brug (2024), for example, find that our neighbours' political preferences come to shape ours only if we feel strongly embedded in our local community. Perhaps, then, the null overall effects reported for both 'mover types' in our study conceal that those who form particularly tight knit bonds in their new communities might experience some assimilation affects. Another possibility is that we don't find evidence of contextual effects because different kinds of effects cancel one another out. If some people react to the greater diversity typically experienced in larger and more densely populated settlements by becoming more pro-immigration (as scholars including Jolly and DiGuisto (2014) have shown), and others react by becoming more anti-immigration (as Igarashi and Laurence (2021) have shown), then it could be that the overall effect of moving to more populous settlements on immigration nets out at zero simply because it combined two opposite non-zero effects of approximately equal strength. That different kinds of people might respond to the same environmental factors in different ways would go some way in explaining the mixed body of evidence that has been amassed on how exposure to diversity shapes individual attitudes in existing work. Both possibilities discussed here lead to the same recommendation.

Future studies should be more attentive to individual differences in the shaping effects of contextual influences.

In showing that people who live in larger and more densely populated places do indeed think differently to those who live in smaller and less densely populated areas, but that this patterning is more complex than the simple urban-rural divide in politics to which many existing studies have alluded[7], this study has important implications for those interested in understanding the spatial patterning of politics. We find that there is a relatively linear association between settlement size and immigration attitudes in Britain, with the populations of increasingly large settlement types tending to be more pro-immigration than those of each smaller settlement type, except for individuals living in large settlements who are, on average, the most pro-immigration of all – being even more so than those living in major settlements. Perhaps even more interestingly, we find that all these geographical differences in immigration attitudes can be explained away by controlling for population composition, except those observed between large and minor settlements. Clearly then, the complexity of this geographical divide over attitudes, and the questions of *how* and *why* this came to be, cannot be properly addressed through statistical analyses that rely on the simple binary classifications of the urban-rural divide that have been adopted in much of the existing research in this space. Future studies should use more nuanced definitions of urbanity and rurality, which allow for finer-grained distinctions between settlements of different sizes and/or densities, and explore the extent to which the same patterns detected here hold up in different contexts and for different types of attitudes.

To conclude, this paper clearly demonstrates that more populous places are more cosmopolitan than less populous places, and that though this relationship is not perfectly

[7] For notable exceptions, see Luca et al (2023), Hesstvedt and Saglie (2025) and Maxwell (2019).

linear (with residents of large settlements holding more cosmopolitan attitudes than those of major settlements, on average) it is certainly more linear than many existing studies have implied. We present robust empirical evidence to show that *compositional* effects play a much stronger role in determining this spatial patterning than do *contextual* effects. Like Maxwell (2019) we thus posit that the geographical divides we observe over politics are second-order manifestations of deeper demographic and cultural divides. More populous places are more 'cosmopolitan' primarily because people who hold specific kinds of attitudinal profiles, and who have the kinds of socio-demographic characteristics that are associated with these, are more likely to choose to live these places and because people who do not choose to live elsewhere. Not because there is something distinctive about the experience of living in more populous places. Though the existence of such contextual effects should not yet be ruled out completely and may matter at the margins. Further studies of individual differences in 'average' contextual effects are needed to assess this possibility.

Data Availability Statement: should this article be accepted for publication at the British Journal of Political Science the code required to replicate our analysis will be published at the journal's Dataverse site. The data needed to replicate our analysis can be downloaded by registered UK Data Service users at the following link: https://datacatalogue.ukdataservice.ac.uk/studies/study/8202#details

Competing Interests: the authors declare none.

Funding Declaration: the authors received no funding to assist with the research presented in this manuscript.

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Supplementary Materials for: Context or Composition? Explaining the Geographical Polarisation of Cosmopolitan Attitudes in Britain.

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1 Classifying Local Authorities by Settlement Size

To classify local authorities by size or degree of urbanity, we avoid using local authority-level population data. Local authorities are administrative geographies that do not necessarily mirror the geographic boundaries of the places that people inhabit and move between, so we instead adopt an approach that incorporates data from the geographic units in which people live. The 2011 Census of England and Wales involved the creation of a new statistical geography - the Built Up Area (henceforth BUA) - which is designed to capture the precise contours of the 6000 towns, cities and villages across England and Wales in which the vast majority of people live (ONS Geography, 2015). We take BUA-level population data as the starting point for our classification, which we believe facilitates a more accurate measure of population size.

Taking population figures for each BUA from the 2011 Census, we then identify the local authorities in which each BUA sits (BUAs tend to be much smaller than local authorities, with most local authorities comprised of several different built-up areas). We then compute the weighted average of BUA population size across the BUAs that sit within each local authority. This provides us with a population/urbanity measure that is derived from—and thus better captures—the size of the places where people actually live, rather than a arbitrary figure based on administrative boundaries.

We employ a weighting scheme to account for the fact that not all BUAs sit wholly within a single local authority. BUAs which sit completely within one local authority are given a weight of 1; BUAs which sit only partially within a local authority are given a weight of $1/n$, where n is the number of built up areas that sit 'partially' within that local authority. We then take the grouped average - at the local authority level - of these weighted BUA population figures. We use the median average population size due to concerns about the potential sensitivity of the mean measure to outliers (i.e. an extremely large BUA could drag up the local authority population average substantially).

We then classify local authorities into population groups using these BUA population averages, following the BUA classification scheme provide by the UK Office for National Statistics (ONS Geography, 2013). Specifically, local authorities with an average BUA population of less than 10,000 are classified as minor; local authorities with an average BUA population between 10,000 and 99,999 are classified as small; local authorities with an average BUA population between 100,000 and 499,999 are classified as medium; local authorities with an average BUA population between 500,000 and 999,999 are classified as large; and local authorities with an average BUA population greater than 1m are classified as major.

Table 1.1 displays the number of local authorities within each category. The majority of local authorities are minor, with major local authorities the second-most frequent.

Table 1.1: Local Authorities Classified by BUA Population Averages

	Minor	Small	Medium	Large	Major
N	230	24	38	7	49
%	66.09	6.90	10.92	2.01	14.08

Figure 1.1 provides this same information in graphical form, mapping local authorities in England and Wales and coding them by category. As we would expect, major local authorities are clustered in urban centres like London, Manchester and Birmingham. A small number of local authorities are missing from this map due to local authority restructuring over the study period.

Figure 1.1: Local Authorities by BUA Population Averages



2 Compositional Analysis

2.1 Attitudinal Differences Between Locality Types

Table 2.1: The Effect of Locality Type and Socio-Demographics on Immigration Attitudes

VARIABLES	**Model I**	**Model II**
LA Type		
Major	0.544***	0.105
	(0.132)	(0.087)
Large	0.749***	0.504***
	(0.222)	(0.177)
Medium	-0.012	-0.035
	(0.033)	(0.062)
Small	0.073	0.033
	(0.134)	(0.105)
Age (years)		-0.028***
		(0.001)
Ethnicity (white vs non-white)		-0.856***
		(0.052)
UK-born (vs otherwise)		-0.244***
		(0.044)
Degree-educated (vs otherwise)		1.158***
		(0.027)
Occupational class		
Managerial and professional		0.769***
		(0.032)
Intermediate		0.350***
		(0.035)
Small employers		0.430***
		(0.047)
Lower supervisory/technical		0.042
		(0.043)
Wave FE	YES	YES
Observations	341,437	341,437

Notes: Results are OLS regression coefficients, with robust SEs (clustered by individual) in parentheses. The reference category for LA type is *Minor*; for occupational groups it is *Routine/Semi-Routine Occupations*. ***$p < 0.001$.

2.2 Sorting Analysis

Below, we replicate the model presented in table 3 of the manuscript, but by fitting an ordered logistic regression using the categorical version of the local authority classification as the outcome variable. Two-way clustering of standard errors is implemented with the vcemway package in Stata (Gu and Yoo, 2019).

Results are presented below in Table 2.2, and show that results mirror those of the OLS specification with population counts used as the outcome variable. The only point of difference is that whilst education predicts moving in the OLS setup, it does not predict moving when this categorical DV is used. Pre-move immigration attitudes, age and ethnicity all predict moving in the expected direction; people who are more supportive of immigration are less likely to move to smaller localities, whilst older respondents and white respondents are more likely to move to smaller localities.

Table 2.2: The Effect of Socio-demographic Characteristics and Immigration Attitudes on 'Move Type', ordered logit replication

VARIABLES	
Immigration attitudes T-1	-0.034* (0.015)
Age T-1	0.017*** (0.003)
White T-1	0.558*** (0.103)
UK Born T-1	-0.093 (0.107)
Degree-educated T-1	-0.100 (0.065)
Managers and Professionals T-1	0.127 (0.096)
Intermediate T-1	0.192 (0.118)
Small employers T-1	0.136 (0.152)
Lower supervisory/technical T-1	0.155 (0.160)
Pre-move LA: Large	0.249 (0.266)
Pre-move LA: Major	-1.116*** (0.159)
Pre-move LA: Medium	0.088 (0.119)
Pre-move LA: Small	-0.275+ (0.158)
Wave FE	YES
Observations	5,002

Notes: Results are raw ordered logit coefficients, with robust SEs (clustered by individual and post-move local authority) in parentheses. The reference category for pre-move LA type is *Minor*; for occupational groups it is *Routine/Semi-Routine Occupations*. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

3 Contextual Effects - Further Analysis

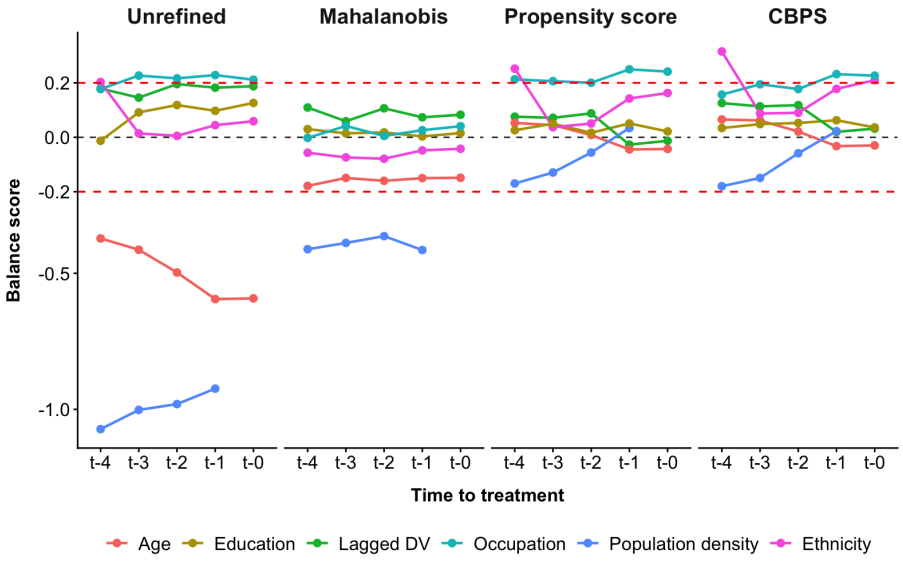
3.1 Large Moves

One possible explanation for the absence of contextual effects in our main PanelMatch specification is that we include all moves that involve a change in density-percentile. Our treatment estimates could thus be biased downward by the inclusion of moves that represent only a minor change in population density.

To address this concern, we rank moves to both higher-density and lower-density MSOAs based on the gap between the density-percentile at T_{-1} and the density-percentile at T. For both moves to higher-density and lower-density MSOAs, we use this ranking to limit our definition of treated units to the largest 50% of moves by percentile difference. This leaves us with 531 treated units who move to higher-density MSOAs and 607 treated units who move to lower-density MSOAs. The moves made by these two groups constitute a meaningful change in residential context. Among those who move to higher-density MSOAs, the average (mean) increase in density-percentile (from MSOA at T_{-1} to MSOA at T) is 39.64; among those who move to lower-density MSOAs, the average decrease in density-percentile is 44.93.

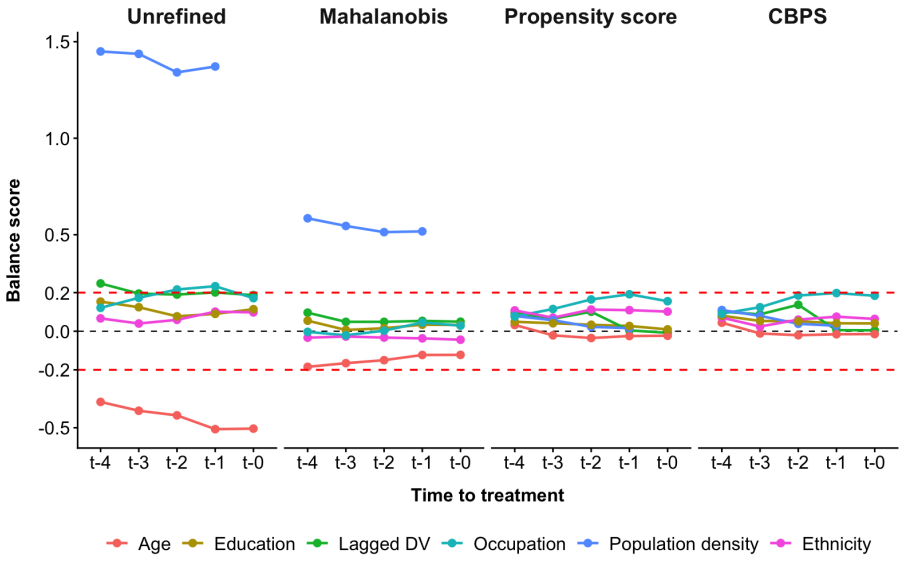
As with the main PanelMatch specification, we construct unrefined candidate sets using 4-wave lags. The median candidate set for higher-density treated units contains 2190 control units (minimum = 155), whilst the median candidate set for lower-density treated units is 2196 (minimum = 152). Refining on ethnicity, age, education, occupation, pre-treatment population density, only propensity score and CBPS matching achieve acceptable degrees of balance, as shown in Figure 3.1a and Figure 3.1b.

Figure 3.1a: Balance Scores, Large Moves to Higher Density MSOAs



Notes: The red dashed lines indicate the ± 0.2 standard deviation boundary used as a cut-off for acceptable balance.

Figure 3.1b: Balance Scores, Large Moves to Lower Density MSOAs



Notes: The red dashed lines indicate the ± 0.2 standard deviation boundary used as a cut-off for acceptable balance.

We thus estimate the ATTs for moving to higher-density and lower-density MSOAs among the propensity score and CBPS refined sets. Results are presented below in Table 3.1

Table 3.1: ATTs for Moving to Higher-Density and Lower-Density MSOAs on Immigration Attitudes

Matching	**Move to higher-density MSOA**				**Move to lower-density MSOA**			
	ATT	**SE**	**Lower CI**	**Upper CI**	**ATT**	**SE**	**Lower CI**	**Upper CI**
PS	0.04	0.081	-0.125	0.199	-0.038	0.078	-0.189	0.11
CBPS	0.033	0.078	-0.111	0.186	-0.004	0.074	-0.154	0.139

Notes: ATT = average treatment effect on the treated. CI = bootstrap confidence interval (1000 repetitions).

3.2 Alternative Covariate Specification

In our main PanelMatch specification, we match on T_0 education and occupation. This was done to separate out the effects of changing residential context from the effects of changes in education level or occupation, which often co-occur with changes in residential location (i.e. moving to a large urban centre for a new job). However, as these covariates are also potentially post-treatment to changing residential context, matching on them at T_0 may be overly conservative.

To confirm that the decision to match on T_0 occupation and education is not behind our null results, we re-estimate the PanelMatch procedure, but alter the refinement procedure so that we match on occupation and education at T_{-4} to T_{-1} only. All other aspects of the estimation procedure remain the same.

To mirror the main model specification, we derive ATT estimates from matched sets refined using Mahalanobis and CBPS. Results are presented below in Table 3.2.

Table 3.2: ATTs for Moving to Higher-Density and Lower-Density MSOAs on Immigration Attitudes: No Contemporaneous Occupation and Education

Matching	**Moving to higher-density MSOA**				**Moving to lower-density MSOA**			
	ATT	**SE**	**Lower CI**	**Upper CI**	**ATT**	**SE**	**Lower CI**	**Upper CI**
Mahalanobis	0.012	0.057	-0.101	0.126	-0.008	0.054	-0.113	0.089
CBPS	0.092	0.057	-0.016	0.208	0.016	0.054	-0.09	0.127

Notes: ATT = average treatment effect on the treated. CI = bootstrap confidence interval (1000 repetitions).

3.3 Impressionable Years Analysis

Our main PanelMatch specification imposes the assumption of homogeneous treatment effects, or that the impact of changing residential context on immigration attitudes will be the same across the treatment population. If this assumption is incorrect, our main model specification could be obscuring a treatment effect among subgroups.

One theory which provides reason to question the assumption of homogeneous treatment effects is the impressionable years hypothesis (Sears and Funk, 1999). The impressionable years hypothesis argues that political attitudes and orientations are most malleable and changeable during the period of adolescence and early adulthood, where limited experience with the social and political world makes individuals receptive to socializing influences from people, places and political events (Sears and Valentino, 1997; Jennings and Markus, 1984). The impressionable years hypothesis implies that, in our research context, individuals who experience a change in residential context during their impressionable years will be more receptive to contextual effects - and thus more likely to update their immigration attitudes - than those who move later in life.

To assess this possibility, we compare the effect of moving to higher-density and lower-density MSOAs exists among individuals who move at age 30 or under vs those who move later in life. Results are presented in Table 3.3. Whilst the point estimates for moving to higher-density MSOAs are larger among movers aged 30 and under, consistent with the logic of the impressionable years hypothesis, we find no evidence of a treatment effect across any of these models.

Table 3.3: ATTs for Moving to Higher-Density and Lower-Density MSOAs on Immigration Attitudes: Impressionable Years

Matching	**Age group**	**Higher-density MSOA**				**Lower-density MSOA**			
		ATT	**SE**	**L. CI**	**U. CI**	**ATT**	**SE**	**L. CI**	**U. CI**
Mahalanobis	≤ 30	0.125	0.106	-0.078	0.334	-0.028	0.116	-0.261	0.189
	> 30	-0.019	0.069	-0.159	0.111	-0.019	0.058	-0.132	0.092
CBPS	≤ 30	0.134	0.106	-0.075	0.344	-0.031	0.115	-0.206	0.25
	> 30	0.03	0.074	-0.108	0.177	0.006	0.61	-0.122	0.119

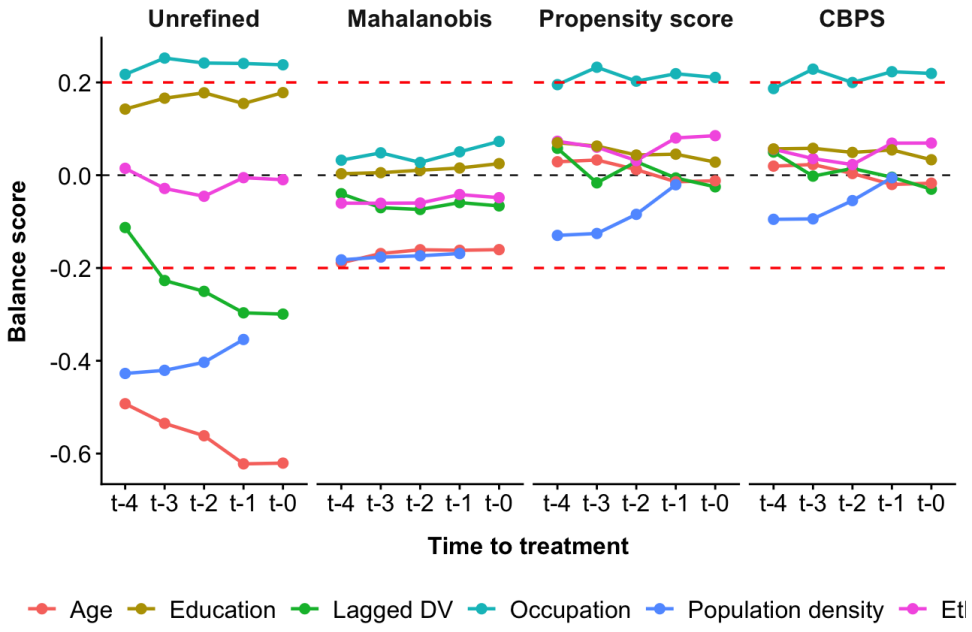
Notes: ATT = average treatment effect on the treated. CI = bootstrap confidence interval (1000 repetitions).

3.4 An Alternative Outcome Measure

Next, we test whether moving to higher-density and lower-density MSOAs has an effect on an alternative (but related) outcome measure. We focus on attitudes to European Integration, measured on a 0-10 scale (0 = 'Unite fully with the European Union'; 10 = 'Protect our independence'). This variable is measured at all waves from 1-30, apart from wave 5. Whilst the issues of European integration and immigration are closely related, the former was arguably more salient for much of the period covered by the BESIP, given the importance of the Brexit referendum and its aftermath during this period. A focus on attitudes to European integration might thus constitute an easier test of the contextual effects hypothesis.

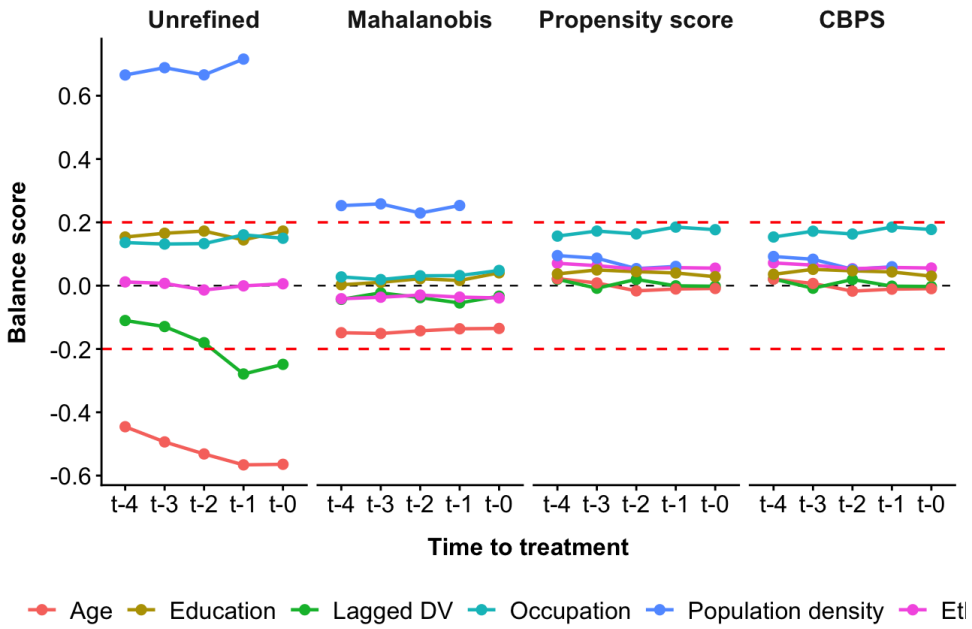
For these models, we follow the PanelMatch specification used in the main models, with the only alteration being the outcome measure. The number of treated units in these models are slightly larger than for immigration attitudes - 1300 individuals move to higher-density MSOAs in the data, and 1648 individuals move to lower-density MSOAs. Figure 3.4a and Figure 3.4b display balance scores for each covariate from T_{-4} to T_0.

Figure 3.4a: Balance Scores, Moving to Higher-Density MSOAs on EU Atitudes



Notes: The red dashed lines indicate the ± 0.2 standard deviation boundary used as a cut-off for acceptable balance.

Figure 3.4b: Balance Scores, Moving to Lower-Density MSOAs on EU Attitudes



Notes: The red dashed lines indicate the ± 0.2 standard deviation boundary used as a cut-off for acceptable balance.

Mahalanobis matching achieves best balance for moving to higher-density MSOAs, whilst propensity score and CBPS matching achieve best balance for moving to lower-density MSOAs. We thus estimate the ATTs for moving to higher-density and lower-density MSOAs across all three refinement procedures. Results are presented below in Table 3.4. We find no evidence of an effect of moving to higher-density or lower-density MSOAs on attitudes to European integration.

Table 3.4: ATTs for Moving to Higher-Density and Lower-Density MSOAs on Attitudes to European Integration

Matching	**Move to higher-density MSOA**				**Move to lower-density MSOA**			
	ATT	**SE**	**Lower CI**	**Upper CI**	**ATT**	**SE**	**Lower CI**	**Upper CI**
Mahalanobis	-0.027	0.056	-0.139	0.083	0.077	0.048	-0.013	0.169
PS	-0.071	0.059	-0.184	0.046	-0.005	0.048	-0.097	0.089
CBPS	-0.095	0.057	-0.204	0.013	-0.004	0.048	-0.096	0.088

Notes: ATT = average treatment effect on the treated. CI = bootstrap confidence interval (1000 repetitions).

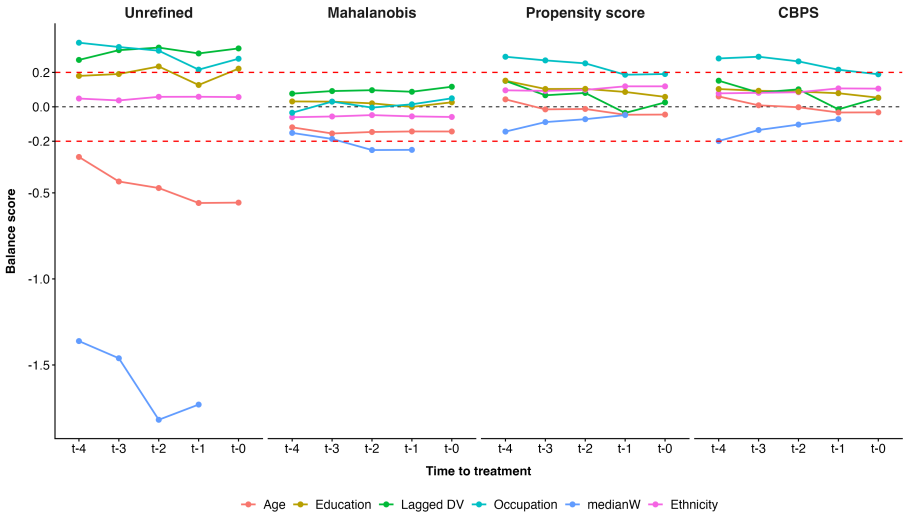
3.5 Local Authority Tests

The BESIP data provides us with local authority identifiers for each survey respondent at each wave. We define moves as instances where the local authority in which an individual resides at timepoint T is different to at T_{-1}. We then split these moves into two types and estimate the effect of each separately: moves to more populous settlements (when a respondent resides in a local authority at T that has a larger average settlement size than at T_{-1}); and moves to a less populous settlement (when a respondent resides in a local authority at T that has a smaller average settlement size than at T_{-1}).

We construct matched sets with 4 lags (i.e., eventually treated units are matched with control units that share treatment and covariate histories for 4 waves prior to treatment). This results in 609 treated units who move to more populous settlements; and 912 treated units who move to less populous settlements. The median candidate set for treated units who move to for more populous LAs contains 1785 control units (minimum = 161), whilst the median candidate set for treated units who move to less populous LAs is 2233 (minimum = 157).

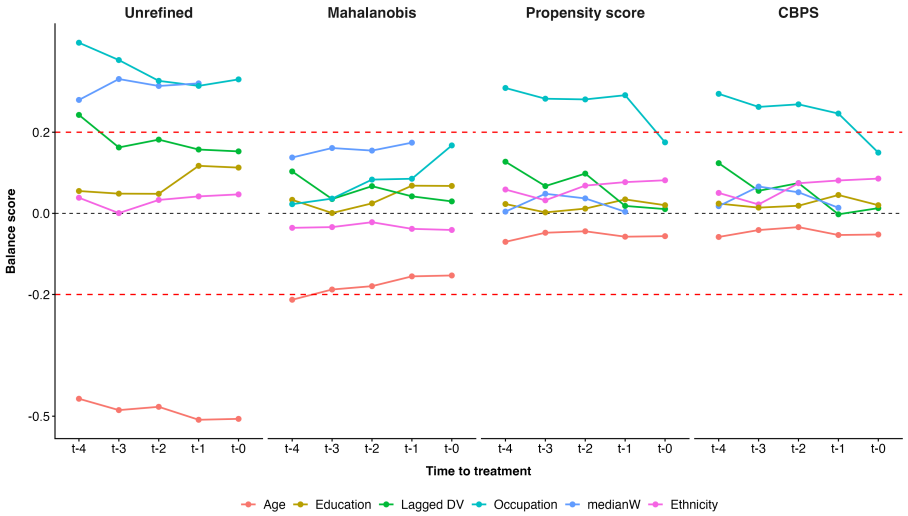
We refine the matched sets by matching on the following covariates: ethnicity (a dummy for whether the respondent is white), age, education (a dummy for whether the respondent has a degree, from T_0 to T_{-4}), occupation (a dummy for whether the respondent works in a managerial or professional occupation, from T_0 to T_{-4}), pre-treatment population size (from T_{-1} to T_{-4}) and the lagged outcome (from T_{-1} to T_{-4}). We use three refinement procedures: Mahalanobis distance matching, propensity score matching, and covariate balancing propensity score matching. As shown in Figure 3.5a and Figure 3.5b below, Mahalanobis distance matching achieves the best covariate balance on all covariates across both moving to more and less populous settlements (using ± 0.2 SD as the cut-off for acceptable balance, as suggested by Rauh, Kim and Imai, 2025).

Figure 3.5a: Balance Scores, Moving to Larger Local Authorities



Notes: The red dashed lines indicate the ± 0.2 standard deviation boundary used as a cut-off for acceptable balance.

Figure 3.5b: Balance Scores, Moving to Larger Local Authorities



Notes: The red dashed lines indicate the ± 0.2 standard deviation boundary used as a cut-off for acceptable balance.

Given that Mahalanobis distance matching achieves the best covariate balance, we estimate the ATT from the matched sets using this refinement strategy. As with the MSOA models, we estimate bootstrap confidence intervals (1000 reps). Results are presented below in Table 3.5. The ATT of moving to a less populous local authority is null, consistent with the MSOA results. Results do suggest that moving to a more populous local authority has a positive effect on immigration attitudes - however, given that the lower CI crosses zero and the MSOA results are non-significant, we do not place much weight on this result.

Table 3.5: ATTs for Moving to More and Less Populous Settlements on Immigration Attitudes

Matching	**Moving to more populous LA**				**Moving to less populous LA**			
	ATT	**SE**	**Lower CI**	**Upper CI**	**ATT**	**SE**	**Lower CI**	**Upper CI**
Mahalanobis	0.12	0.07	-0.01	0.27	-0.02	0.06	-0.139	0.10

Notes: ATT = average treatment effect on the treated. CI = bootstrap confidence interval (1000 repetitions).

3.6 TWFE Analysis

Whilst we believe the matching approach used in our main specification is the best available strategy for testing the effects of changing residential context, testing across different estimation strategies is important to validate that the lack of evidence for contextual effects is not the artefact of a specific estimator or modelling strategy. Given this, we conduct an additional test of the effect of changing residential context on immigration attitudes using a different estimation strategy.

Specifically, we adopt the two-way fixed effects (henceforth TWFE) strategy used in previous tests of contextual effects (Maxwell, 2019; Maxwell, 2020), which again leverages changes in residential context to identify contextual effects. Using the same BESIP data as our previous analysis, we again focus on changes in MSOAs to define moves. Specifically, we define a person-observation at wave T as a move to a higher-density MSOA when the individual resided in a different MSOA with a lower density percentile at T_{-1}; moves to lower-density MSOAs are defined as person-observations at T when the individual resided in a different MSOA with a higher density percentile at T_{-1}.

As with our matching-based models, we test the effect of moving to higher-density MSOAs in separate models to the effect of from moving to lower-density MSOAs. In each model, we regress immigration attitudes on three explanatory variables: a dummy which equals 1 for all person-observations when a move occurs (i.e. when the MSOA at T is different to the MSOA at T_{-1}) and coded 0 otherwise; a person- and wave-specific integer variable whose value corresponds to the number of waves before a move occurs; and a person- and wave-specific integer variable which corresponds to the number of waves after a move occurs. We treat these latter two variables as categorical. We also control for age, occupational class and education.

These models include individual and wave fixed effects. As such, the coefficients represent the within-person effect of changing residential context on immigration attitudes, accounting for time-invariant individual unobservables and common time trends. These models effectively estimate the average difference in immigration attitudes between movers and non-movers over time, both before and after moving. As such, they enable us to compare the attitudinal time trend for movers vs non-movers and assess whether movers update their attitudes following a change in residential context. Evidence that the trend for immigration attitudes among movers begins to diverge after moving would support the contextual effects hypothesis (indicating that movers begin to be influenced by their new residential context following this change).

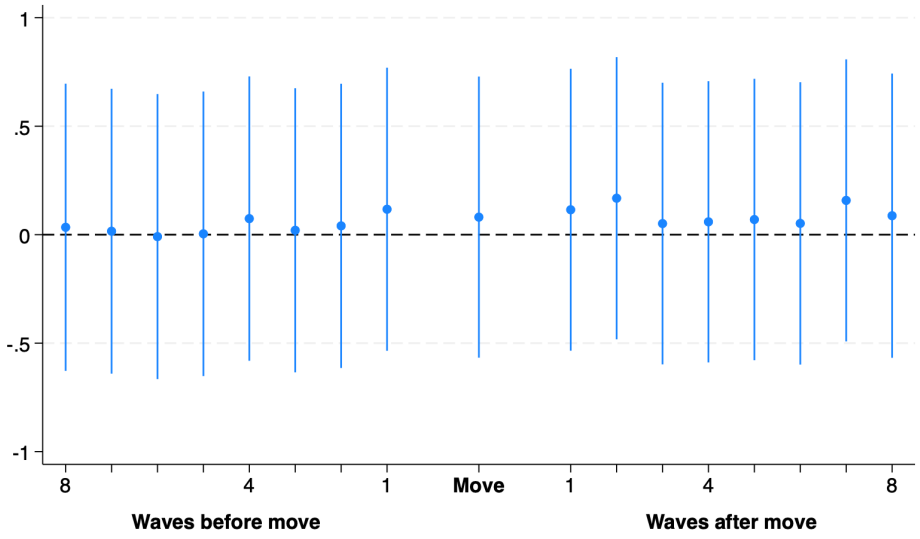
This estimation strategy has three advantages for identifying contextual effects and disentangling them from selection effects. First, the inclusion of time and individual fixed effects means we account for both common shocks and time-invariant individual differences that might influence both immigration attitudes and the decision to move. Second, given that any potential contextual effects of moving to a more/less populous area are unlikely to be instantaneous but accumulate over time, the post-move indicator allows us to capture any lagged effects of moving. Third, the inclusion of a pre-move indicator enables us to identify whether individual attitudes begin to change prior to moving, which would be consistent with the logic of selection effects.

We begin by testing models for moves to higher-density MSOAs (there are 2323 movers to

higher-density MSOAs contained in this model). As this model involves a large number of coefficients – one for each wave pre-move and post-move plus a coefficient for the move wave – we plot these coefficients below in Figure 3.6a. We only plot before and after coefficients that contain at observations for 500+ movers. The horizontal line at y = 0 represents the time-trend in immigration attitudes for people who do not move, and each coefficient represents the difference in attitudes among movers at each wave (relative to the trend among non-movers), along with 95% confidence intervals.

If a change of residential context influences immigration attitudes, we would expect to see the coefficients deviate from zero in the waves after a move occurred, but not before. However, Figure 3.6a shows that immigration attitudes among movers do not deviate from zero before or after moving (the confidence intervals cross zero at all waves), providing no evidence that moving to higher-density MSOAs leads individuals to adopt more favourable attitudes toward immigration.

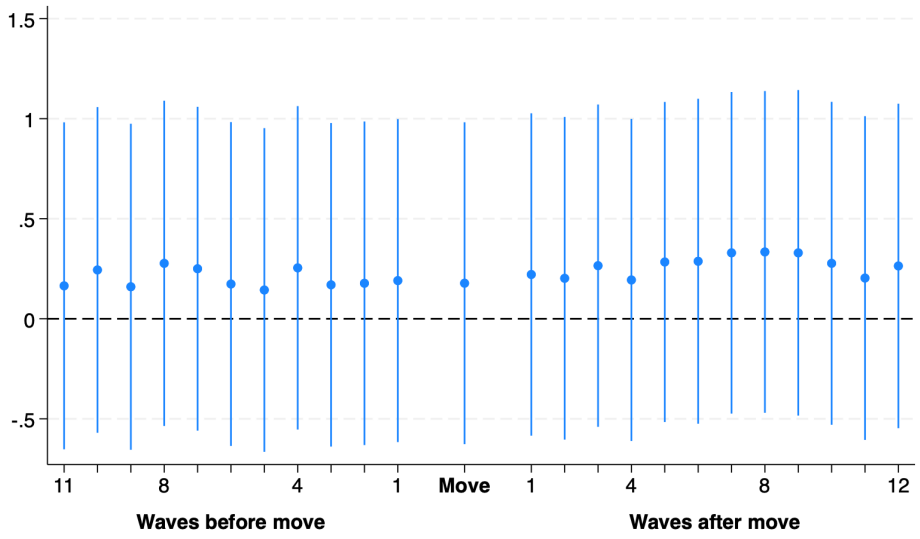
Figure 3.6a: Immigration Attitude Time Trends for Moving to Higher-Density MSOAs



Notes: Coefficients (surrounded by 95% confidence intervals) from OLS regression models with person and time fixed effects. The x-axis plots waves before and after the move. N = 316,201.

Next, we test whether moving to lower-density MSOAs impacts immigration attitudes, and plot the coefficients below in Figure 3.6b. Again, we only plot before and after coefficients that contain at observations for 500+ movers. This model contains 3163 movers to lower-density MSOAs. We again find no evidence to support the contextual effects hypothesis, with the coefficients not deviating from zero after moving.

Figure 3.6b: Immigration Attitude Time Trends for Moving to Lower-Density MSOAs



Notes: Coefficients (surrounded by 95% confidence intervals) from OLS regression models with person and time fixed effects. The x-axis plots waves before and after the move. N = 316,201.

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